

After the classes are defined, we need to create the migration file.

Migration

Migration is to a database what GIT or SVN is to the code base. It keeps track of changes in the schema by creating files for each one. For each change, we need to create a migration file and then execute this file to replicate the specified change to the underlying database. The commands for these operations vary based on the ORM we use.

Why ORM is needed

While developing any application, by using ORM, we get the following advantages.

1. Productivity
 - Eliminates lots of repetitive code
 - Database schema is generated automatically
 2. Maintainability
 - Fewer lines of code – easier to understand
 - Easier to manage changes in the table/class
 3. Performance
 - Lazy loading – associations are fetched when needed
 - Caching
 4. Vendor-independent
 - The underlying database is abstracted away
- In the code below, ORM supports multiple databases. A

query in ORM looks like what follows:

```
Employee.objects.filter(first_name='xxxx')
```

...which is the same as what's shown below:

```
select * from employee where first_name='xxxx'
```

In the above fashion, any SQL query can be written in ORM. There is no need for the developer to write complex SQL queries. An ORM query is converted into an equivalent vendor-specific database query for execution. This is a lazy way of doing things, i.e unless we try to retrieve some properties from the query result set, the conversion process or the query execution won't happen.

Django has its own built-in ORM. We can also use SQLAlchemy, which is an open source ORM toolkit for the Python frameworks. Apart from these two, there are some other open source ORMs such as SQLAlchemy, Propel, Redbean, and so on.

To learn more about the ORM in Django, go to <https://docs.djangoproject.com/en/2.2/topics/db/models/>. **END** 🐧

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